ASSIGNMENT 8

Basic SQL Queries and Aggregate Functions

HOSPITAL MANAGEMENT SYSTEM

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Course: BCA (Hons)

Subject: Database Management System

1. Objective

Demonstrate WHERE filtering, ORDER BY sorting, GROUP BY grouping, HAVING group filtering and COUNT, SUM, AVG, MIN and MAX using the sample data from Assignment 7.

Query 1 — WHERE: completed appointments

```
SELECT appointment_id, patient_id, doctor_id, appointment_date, status
FROM Appointments
WHERE status = 'Completed';
```

Expected output / explanation: Expected rows: appointments 101 and 102. WHERE filters individual rows before grouping.

Query 2 — ORDER BY: bills from highest to lowest

```
SELECT bill_id, total_amount, payment_status
FROM Bills
ORDER BY total_amount DESC;
```

Expected output / explanation: Expected order: 401 (850.00), 403 (750.00), 402 (600.00), 405 (500.00), 404 (200.00).

Query 3 — GROUP BY and COUNT: appointments per doctor

```
SELECT doctor_id, COUNT(*) AS appointment_count
FROM Appointments
GROUP BY doctor_id
ORDER BY doctor_id;
```

Expected output / explanation: Expected: doctor 1 = 2, doctor 2 = 3, doctor 3 = 1.

Query 4 — HAVING: doctors with more than one appointment

```
SELECT doctor_id, COUNT(*) AS appointment_count
FROM Appointments
GROUP BY doctor_id
HAVING COUNT(*) > 1;
```

Expected output / explanation: Expected: doctor 1 = 2 and doctor 2 = 3. HAVING filters groups after aggregation.

Query 5 — SUM: total billed amount by payment status

```
SELECT payment_status, SUM(total_amount) AS total_amount
FROM Bills
```

```
ORDER BY payment_status;
```

Expected output / explanation: Expected: Paid = 850.00; Partially Paid = 600.00; Pending = 1250.00; Refunded = 200.00.

Query 6 — AVG, MIN, MAX: bill summary

```
SELECT COUNT(*) AS bill_count,
       SUM(total_amount) AS total_billed,
       ROUND(AVG(total_amount), 2) AS average_bill,
       MIN(total_amount) AS smallest_bill,
       MAX(total_amount) AS largest_bill
FROM Bills;
```

Expected output / explanation: Expected: 5 bills; total 2900.00; average 580.00; minimum 200.00; maximum 850.00.

Query 7 — WHERE with date range

```
SELECT appointment_id, patient_id, appointment_date, appointment_time
FROM Appointments
WHERE appointment_date BETWEEN '2026-10-20' AND '2026-10-22'
ORDER BY appointment_date, appointment_time;
```

Expected output / explanation: Expected: appointments 101, 102, 103 and 104.

Query 8 — medicine price aggregate

```
SELECT COUNT(*) AS medicine_count,
       ROUND(AVG(price), 2) AS average_price,
       MIN(price) AS lowest_price,
       MAX(price) AS highest_price
FROM Medicines;
```

Expected output / explanation: Expected: 5 medicines; average price 3.90; minimum 2.50; maximum 6.00.

Conclusion

These queries cover the required filtering, sorting, grouping, HAVING and aggregate functions. Output descriptions assume Assignment 7 sample data was inserted successfully into an empty database.

ASSIGNMENT 8 — OUTPUTS AND VERIFICATION

Query 1 — WHERE status = 'Completed'

appointment_id	patient_id	doctor_id	appointment_date	status
101	1	1	2026-10-20	Completed
102	2	2	2026-10-20	Completed

Query 2 — ORDER BY bill total DESC

bill_id	total_amount	payment_status
401	850.00	Paid
403	750.00	Pending
402	600.00	Partially Paid
405	500.00	Pending
404	200.00	Refunded

Query 3 — GROUP BY doctor, COUNT appointments

doctor_id	appointment_count
1	2
2	3
3	1

Query 4 — HAVING COUNT(*) > 1

doctor_id	appointment_count
1	2
2	3

Query 5 — SUM by payment status

payment_status	total_amount
Paid	850.00
Partially Paid	600.00
Pending	1250.00
Refunded	200.00

Query 6 — Bill aggregates

bill_count	total_billed	average_bill	smallest_bill	largest_bill
5	2900.00	580.00	200.00	850.00

Query 7 — Appointments between 2026-10-20 and 2026-10-22

appointment_id	patient_id	appointment_date	appointment_time
101	1	2026-10-20	09:00:00
102	2	2026-10-20	10:30:00

103	3	2026-10-21	11:00:00

appointment_id	patient_id	appointment_date	appointment_time
104	4	2026-10-22	14:00:00

Query 8 — Medicine price aggregates

medicine_count	average_price	lowest_price	highest_price
5	3.90	2.50	6.00